

Translated

From: e-maxx.ru

Euler's totient function

Euler's totient function, also known as **phi-function** $\phi(n)$, counts the number of integers between 1 and n inclusive, which are coprime to n . Two numbers are coprime if their greatest common divisor equals 1 (1 is considered to be coprime to any number).

Here are values of $\phi(n)$ for the first few positive integers:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
$\phi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8	8	16	6	18	8	12

Properties

The following properties of Euler totient function are sufficient to calculate it for any number:

- If p is a prime number, then $\gcd(p, q) = 1$ for all $1 \leq q < p$. Therefore we have:

$$\phi(p) = p - 1.$$

- If p is a prime number and $k \geq 1$, then there are exactly p^k / p numbers between 1 and p^k that are divisible by p . Which gives us:

$$\phi(p^k) = p^k - p^{k-1}.$$

- If a and b are relatively prime, then:

$$\phi(ab) = \phi(a) \cdot \phi(b).$$

This relation is not trivial to see. It follows from the [Chinese remainder theorem](#). The Chinese remainder theorem guarantees, that for each $0 \leq x < a$ and each $0 \leq y < b$, there exists a unique $0 \leq z < ab$ with $z \equiv x \pmod{a}$ and $z \equiv y \pmod{b}$. It's not hard to show that z is coprime to ab if and only if x is coprime to a and y is coprime to b . Therefore the amount of integers coprime to ab is equal to product of the amounts of a and b .

- In general, for not coprime a and b , the equation

$$\phi(ab) = \phi(a) \cdot \phi(b) \cdot \frac{d}{\phi(d)}$$